Data Structures

CSL202

Quiz Set-A

Q1. A library maintains a catalog of book IDs in the form of a binary tree.

The catalog can be uniquely constructed if both the inorder traversal and the preorder traversal of the tree are known.

Given:

Inorder Traversal = [D, B, E, A, F, C]

Preorder Traversal = [A, B, D, E, C, F]

Task:

1. Draw the final tree.

2. After constructing tree write its postorder.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while maintaining the BST property.

Data Structures

CSL202

Quiz Set-A

Q1. A library maintains a catalog of book IDs in the form of a binary tree.

The catalog can be uniquely constructed if both the inorder traversal and the preorder traversal of the tree are known.

Given:

Inorder Traversal = [D, B, E, A, F, C]

Preorder Traversal = [A, B, D, E, C, F]

Task:

1. Draw the final tree.

2. After constructing tree write its postorder.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while maintaining the BST property.

Current tree (course codes inserted in this order): 45, 25, 65, 15, 35, 55, 75, 5,

20, 30, 40

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 25.

Current tree (course codes inserted in this order): 45, 25, 65, 15, 35, 55, 75, 5,

20, 30, 40

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 25.

Data Structures

CSL202

Quiz Set-B

Q1. A college stores student roll numbers in a binary tree for quick searching.

The system can reconstruct the binary tree if the inorder and preorder

sequences are provided.

Given:

Data Structures

CSL202

Quiz Set-B

Q1. A college stores student roll numbers in a binary tree for quick searching.

The system can reconstruct the binary tree if the inorder and preorder

sequences are provided.

Given:

Inorder Traversal = [8, 4, 2, 5, 1, 6, 3, 7]

Preorder Traversal = [1, 2, 4, 8, 5, 3, 6, 7]

Task:

1. Show the final binary tree diagram.

2. After constructing tree write its postorder.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while

maintaining the BST property.

Current tree (course codes inserted in this order): 60, 40, 80, 20, 50, 70, 90

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 80.

Inorder Traversal = [8, 4, 2, 5, 1, 6, 3, 7]

Preorder Traversal = [1, 2, 4, 8, 5, 3, 6, 7]

Task:

1. Show the final binary tree diagram.

2. After constructing tree write its postorder.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while

maintaining the BST property.

Current tree (course codes inserted in this order): 60, 40, 80, 20, 50, 70, 90

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 80.

Data Structures

CSL202

Quiz Set-C

Q1. A school stores roll numbers in a binary tree.

The system can reconstruct the tree if inorder and postorder traversals are known.

Given:

Inorder Traversal = [4, 2, 5, 1, 6, 3, 7]

Postorder Traversal = [4, 5, 2, 6, 7, 3, 1]

Task:

1. Draw the final tree.

2. After constructing tree write its preorder.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while

Data Structures

CSL202

Quiz Set-C

Q1. A school stores roll numbers in a binary tree.

The system can reconstruct the tree if inorder and postorder traversals are known.

Given:

Inorder Traversal = [4, 2, 5, 1, 6, 3, 7]

Postorder Traversal = [4, 5, 2, 6, 7, 3, 1]

Task:

1. Draw the final tree.

2. After constructing tree write its preorder.

maintaining the BST property.

Current tree (course codes inserted in this order): 50, 30, 70, 20, 40, 60, 80

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 50.

Q2. A university stores course codes in a Binary Search Tree (BST).

When a course is discontinued, its code must be deleted from the tree while

maintaining the BST property.

Current tree (course codes inserted in this order): 50, 30, 70, 20, 40, 60, 80

Task:

1. Write the pseudocode to delete a node from a BST.

2. Draw the final tree after deleting node of value 50.